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## **The Continuous-Coefficient Jensen Equation: How Real Mixing Weights Retire Regularity**

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# The Continuous-Coefficient Jensen Equation: How Real Mixing Weights Retire Regularity

**Abstract.** Jensen’s inequality is one of analysis’s workhorse estimates: for every concave function  $G$  and random variable  $\xi$  on an interval,  $\mathbb{E}[G(\xi)] \leq G(\mathbb{E}[\xi])$ . Suppose equality is forced for every two-point distribution—that is,  $p G(u_1) + (1-p) G(u_2) = G(p u_1 + (1-p) u_2)$  for all  $u_1, u_2$  in the interval and all weights  $p \in [0, 1]$ . Must  $G$  be affine?

Yes, and the proof is one line. Substitute  $u_1 = M$ ,  $u_2 = 0$ ,  $p = v/M$  and read  $G(v)$  off the chord through  $(0, G(0))$  and  $(M, G(M))$ —and it needs no regularity hypothesis on  $G$ : no continuity, measurability, monotonicity, or boundedness.

The contrast with rational weights is sharp. The  $p = \frac{1}{2}$  case is equivalent to Cauchy’s additive equation, which admits wildly pathological non-affine solutions (Hamel, 1905) unless a regularity hypothesis intervenes. That historical reflex is useful in the rational setting but misleading here. Classical analysis from Cauchy (1821) to Kestelman (1947) cataloged the minimal tameness conditions that rule the pathology out, and the conditions became standard safeguards in functional-equation arguments—sometimes carried into the continuous-coefficient setting, where the equation itself already excludes the pathology.

This article isolates the asymmetry and makes it explicit. We catalog which classical regularity hypotheses become vestigial under the continuous-coefficient equation, explain why (the Hamel pathology survives rational-weight tests and is exposed at the irrational weights at which the equation now tests  $G$ ), and give a diagnostic separating the saturated-Jensen settings in which the hypotheses are vestigial from the more common settings in which they remain load-bearing. The same five hypotheses appear on both sides of the line: indispensable in the axiomatic characterization of Shannon entropy, vestigial in a calibration computation on an atomless space.

**Keywords:** Jensen equation, Cauchy equation, Hamel basis, functional equation, regularity hypothesis

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**1. INTRODUCTION.** One of the most serviceable inequalities in mathematics—Jensen’s inequality [28],  $\mathbb{E}[G(\xi)] \leq G(\mathbb{E}[\xi])$  for concave  $G$ —raises a natural equality question: when does equality hold for *every* two-point distribution supported on a given interval? The equation expressing universal two-point equality,

$$p G(u_1) + (1-p) G(u_2) = G(p u_1 + (1-p) u_2) \quad (u_1, u_2 \in I, p \in [0, 1]), \quad (\star)$$

imposed on a function  $G: I \rightarrow \mathbb{R}$  on a real interval  $I$ , has at least three forms in classical analysis. They are easy to conflate, and that conflation matters both for the foundational theory and for modern applied arguments in which the equation reappears.

**Three forms of the equation.** The *discrete-coefficient* Jensen equation,

$$G\left(\frac{u_1+u_2}{2}\right) = \frac{G(u_1)+G(u_2)}{2}, \quad (J_2)$$

is the  $p = \frac{1}{2}$  specialization of  $(\star)$ . Setting  $f(x) := G(x) - G(0)$ , the equation  $(J_2)$  is equivalent on a translate of  $I$  to Cauchy's additive equation  $f(x + y) = f(x) + f(y)$ . Cauchy's equation inherits a full apparatus of pathological solutions: without measurability, monotonicity, boundedness on a set of positive Lebesgue measure, or another regularity hypothesis, the equation admits non-affine solutions constructed via a *Hamel basis*—a basis of  $\mathbb{R}$  as a  $\mathbb{Q}$ -vector space, whose existence requires the axiom of choice. The classical pathological-solution apparatus is the work of Hamel [25]; the regularity hypotheses sufficient to recover affineness were cataloged by Cauchy [12], Darboux [14, 15], Sierpiński [45], Kormes [33], Ostrowski [40], and Kestelman [30], with Steinhaus's difference-set theorem [48] supplying the tool behind the modern proofs of the measure-theoretic entries; Reem [41] surveys the catalog.

The *rational-coefficient* Jensen equation,

$$pG(u_1) + (1-p)G(u_2) = G(pu_1 + (1-p)u_2) \\ (u_1, u_2 \in I, \quad p \in [0, 1] \cap \mathbb{Q}), \quad (J_{\mathbb{Q}})$$

strengthens  $(J_2)$  but does not escape its pathology. Cauchy additivity entails  $\mathbb{Q}$ -homogeneity— $f(qx) = qf(x)$  for every  $q \in \mathbb{Q}$  and  $x \in \mathbb{R}$ —by an elementary induction on positive integers, sign reversal for negatives, and the inverse-of-multiplication argument for  $1/n$ . Combining additivity with  $\mathbb{Q}$ -homogeneity returns  $(J_{\mathbb{Q}})$  for  $f$ , hence for  $G$ . So  $(J_{\mathbb{Q}})$ ,  $(J_2)$ , and Cauchy's equation share the same solution class up to constants, and all three inherit the Hamel-basis pathology.

The *continuous-coefficient* form  $(\star)$ , by contrast, is fundamentally different. The strengthening from  $\mathbb{Q}$  to  $\mathbb{R}$  in the range of  $p$  is *genuine*. As we will see,  $(\star)$  does not admit any Hamel pathology: the pathology cannot survive at irrational  $p$ , and the continuous-coefficient equation tests  $G$  at every irrational. That distinction is the subject of the paper.

### The main result.

**Theorem 1.** *Let  $M > 0$  and let  $G: [0, M] \rightarrow \mathbb{R}$  satisfy  $(\star)$  for every  $u_1, u_2 \in [0, M]$  and every  $p \in [0, 1]$ . Then  $G$  is affine on  $[0, M]$ :*

$$G(v) = \frac{G(M) - G(0)}{M} v + G(0), \quad v \in [0, M].$$

The hypothesis list deserves attention: no regularity of any kind—continuity, measurability, monotonicity, or boundedness—is assumed of  $G$ . The proof is one line: a substitution into  $(\star)$  pins  $G(v)$  on the straight line through the endpoints  $(0, G(0))$  and  $(M, G(M))$ . We give it in Section 2 and use the remainder of the paper to explain why the distinction matters.

**The historical detour and the structural moral.** Theorem 1 is elementary. Its substitution technique is folklore in functional equations: Aczél's [1, §2.1.4] contains a dyadic precursor proved (with a continuity hypothesis) by a dyadic-induction argument. The precise statement of Theorem 1—continuous coefficients  $p \in [0, 1]$ , no regularity hypothesis on  $G$ —does not appear in this exact form in the standard references the author consulted; it is recorded here because the recurrence pattern of Section 5 needs it as a separable lemma. The paper's contribution is threefold:

- *The explicit dictionary* (Section 3) of five classical tameness hypotheses on  $G$ , each sufficient to force affineness for  $(J_2)$  but vestigial under  $(\star)$ , together with the structural reason (developed in Section 3).

- *The strict-minimum hypothesis* (Section 4)—the equation  $(\star)$  at a single endpoint configuration already suffices.
- *The diagnostic* (Section 5) for distinguishing saturated-Jensen derivations that genuinely produce the continuous-coefficient equation — where the dictionary’s regularity hypotheses are vestigial — from those that produce a rational-coefficient relative, an additive function into a bounded codomain, a Cauchy relative with a non-affine target, or no real-valued unknown at all (the zeroth question). In the latter cases the same hypotheses — or an existence axiom prior to them — are load-bearing; the misclassification runs in both directions, and we give examples of each.

The article deliberately foregrounds one part of Jensen-equation theory: the regime in which the continuous-coefficient form needs no regularity assumption on the function. Kuczma’s monograph [35] is cited only for the classical regularity catalog, the rational-coefficient pathology, and convex-domain variants, not for the exact continuous-coefficient endpoint-substitution statement isolated here. The substitution that closes  $(\star)$  is one of the simplest observations in that tradition; what justifies writing it down is the dictionary, the structural mechanism, and the diagnostic of applied recurrences.

The classical detour through Hamel and the Polish school answered a perfectly natural question—*what regularity hypothesis suffices for  $(J_2) \Rightarrow \text{affine}$ ?*—but a *different* question from the one that matters for many modern applications. The applied derivation does not produce  $(J_2)$  or  $(J_{\mathbb{Q}})$ ; it produces  $(\star)$  as a saturated identity, and the one-line substitution of Section 2 is the right closure tool. The dictionary of Section 3 makes explicit which classical regularity hypotheses become vestigial in this transition.

This pattern — a continuum built into the model retiring a classical hypothesis — has a distinguished ancestor. In exchange economies with an atomless continuum of traders, Aumann showed that convexity of preferences becomes unnecessary [5], Lyapunov-type convexity supplying for free what the hypothesis used to buy. The present article records the same phenomenon one level down, at the level of a single functional equation on an interval.

**Terminology.** Throughout the paper we refer to the substitution  $u_1 = M$ ,  $u_2 = 0$ ,  $p = v/M$  in  $(\star)$  — by which  $G(v)$  is read off the straight line through the endpoints  $(0, G(0))$  and  $(M, G(M))$  — as the *endpoint substitution*. The descriptive label is our convenience, adopted for brevity in repeated reference rather than as a claim of established terminology.

## 2. THE RESULT AND ITS PROOF.

*Proof of Theorem 1.* Fix  $v \in [0, M]$ . Set

$$u_1 := M, \quad u_2 := 0, \quad p := \frac{v}{M}.$$

The constraints of  $(\star)$  are satisfied:  $u_1 = M \in [0, M]$ ,  $u_2 = 0 \in [0, M]$ , and  $p = v/M \in [0, 1]$  since  $v \in [0, M]$ . Substituting into  $(\star)$ ,

$$\frac{v}{M} G(M) + \left(1 - \frac{v}{M}\right) G(0) = G\left(\frac{v}{M} \cdot M + \left(1 - \frac{v}{M}\right) \cdot 0\right) = G(v).$$

Rearranging yields  $G(v) = G(0) + (G(M) - G(0)) v/M$ . ■

The endpoint substitution pins  $G(v)$  on the straight line from  $(0, G(0))$  to  $(M, G(M))$  for every  $v \in [0, M]$ . Since this determines  $G(v)$  at every  $v$ , the func-

tion  $G$  is fully determined by its values at the two endpoints — it is exactly the affine function joining them.

**Corollary 2.** *Under the hypotheses of Theorem 1,  $G$  is in particular continuous, monotone, Lipschitz, absolutely continuous, and Lebesgue measurable on  $[0, M]$ .*

*Proof.* Every affine function  $G(v) = av + b$  on a bounded interval is differentiable with constant derivative  $G' = a$ , hence continuous; non-decreasing if  $a \geq 0$  and non-increasing if  $a \leq 0$ ; Lipschitz with constant  $|a|$ ; absolutely continuous; and Borel-measurable, hence Lebesgue-measurable. ■

The order of inference matters enough to state explicitly. Under  $(\star)$ , the regularity properties of  $G$  are the *conclusion*, not the hypothesis. Each regularity hypothesis one might be tempted to assume—continuity, monotonicity, boundedness, measurability—is automatically satisfied by any solution of  $(\star)$ . The continuous-coefficient equation supplies its own regularity internally; no external regularity hypothesis is needed.

This is what makes the continuous-coefficient case structurally different from the discrete-coefficient case  $(J_2)$ , where some regularity hypothesis *is* required to rule out the Hamel pathology. We now examine the difference in detail.

### 3. THE DICTIONARY OF VESTIGIAL REGULARITY HYPOTHESES.

**The dictionary.** We place side by side the discrete-coefficient equation  $(J_2)$  and the continuous-coefficient equation  $(\star)$ , and ask which regularity hypotheses on  $G$  classically suffice to force affineness, and whether any such tameness assumption is needed at all. Table 1 lists these hypotheses side-by-side, and Figure 1 draws the comparison as a load diagram: five classical hypotheses, each on its own strong enough to carry the conclusion for  $(J_2)$ , and the same conclusion for  $(\star)$  carried by the bare equation, every strut removed. The caption of Figure 1 records the standard historical attribution of each strut. (For a modern textbook treatment of the regularity catalog for  $(J_2)$  we refer the reader to Kuczma [35, §13.2].)

The ground line carries the point of the figure. With **no hypothesis at all**, the beam falls for  $(J_2)$ : there exists a non-affine, non-measurable, unbounded additive function on  $\mathbb{R}$ —the classical Hamel-basis pathology [25]—that supplies a non-affine solution of  $(J_2)$ . Conversely, **no hypothesis is needed** for  $(\star)$ : Theorem 1’s endpoint substitution closes the proof at the level of the bare functional equation.

The five struts tell the complementary story. For  $(J_2)$ , *some* strut is required; otherwise the Hamel pathology survives. For  $(\star)$ , *none* is required—and yet each is automatically satisfied by any solution (Corollary 2). Each hypothesis is, in the dictionary’s terminology, *vestigial*: classically useful as a sufficient tameness route for  $(J_2)$ , classically expected for  $(\star)$ , but in fact unnecessary for  $(\star)$ .

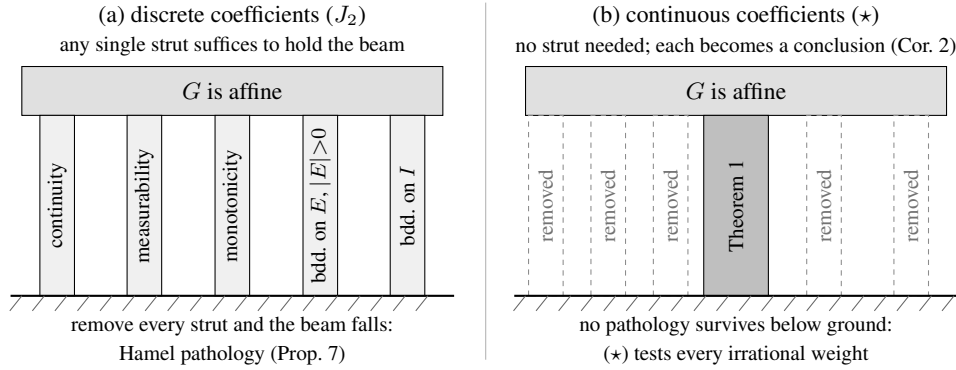
**The structural mechanism.** Why does the Hamel pathology fail to apply to  $(\star)$ ? The mechanism is simple.

A Hamel-basis pathological solution  $G$  of Cauchy’s equation on  $\mathbb{R}$  is  $\mathbb{Q}$ -linear:  $G(0) = 0$  and  $G(qx) = qG(x)$  for every  $q \in \mathbb{Q}$  and  $x \in \mathbb{R}$ . By design,  $G$  is *not*  $\mathbb{R}$ -linear: there exists an irrational  $r \in \mathbb{R}$  and an  $x \in \mathbb{R}$  such that  $G(rx) \neq rG(x)$ .

The straight-line identity in  $(\star)$  at the configuration  $u_1 = M$ ,  $u_2 = 0$  reads

$$pG(M) + (1 - p)G(0) = G(pM). \quad (\star_0)$$

For a Hamel-pathological  $G$  (so  $G(0) = 0$ ),  $(\star_0)$  becomes  $pG(M) = G(pM)$ —exactly the  $\mathbb{R}$ -linearity assertion for  $G$  at the pair  $(M, p)$ . By  $\mathbb{Q}$ -linearity of  $G$ , this



**Figure 1.** The dictionary of vestigial regularity hypotheses, drawn as a load diagram. (a) Under the discrete-coefficient equation ( $J_2$ ), the conclusion “ $G$  is affine” must be carried by a regularity strut, and any single one suffices: continuity (Cauchy [12]), measurability (Sierpiński [45]), monotonicity (Darboux [14]), boundedness on a set of positive measure (Kormes [33]; one-sided bounds suffice, Ostrowski [40], Kestelman [30]), or boundedness on the interval (Darboux [15]). Remove them all and the beam falls: the Hamel-basis pathology [25] supplies a non-affine solution (Proposition 7). Steinhaus’s difference-set theorem [48] powers the modern proofs of the two measure-theoretic struts but is a tool, not a strut; for the historical catalog see Reem [41] and Kuczma [35, §13.2]. (b) Under the continuous-coefficient equation (\*) the bare equation carries the load (Theorem 1); every strut becomes a conclusion rather than a hypothesis (Corollary 2), and no pathology survives below ground, because (\*) tests every irrational weight.

holds for every  $p \in \mathbb{Q}$ . For fixed  $M > 0$ , if the same identity held for every  $p \in [0, 1]$  and  $G$  were additive on  $\mathbb{R}$ , then for any real  $y$  one could choose an integer  $n$  with  $p = |y|/(nM) \leq 1$ ; additivity and sign reversal would give  $G(y) = (y/M)G(M)$ , forcing  $\mathbb{R}$ -linearity. By the failure of  $\mathbb{R}$ -linearity, it *fails* at some irrational  $p$ . So the Hamel-pathological  $G$  satisfies ( $J_{\mathbb{Q}}$ ) (every rational coefficient is OK) but violates (\*) at every irrational  $p$  where the  $\mathbb{R}$ -linearity defect appears.

The structural point fits in one sentence:

**The Hamel pathology survives the rational tests but is exposed by irrational  $p$ ; the continuous-coefficient equation (\*) forecloses on it there.**

Figure 2 draws both halves of that sentence.

The classical program from Cauchy (1821) through Kestelman (1947) can be read, in hindsight, as an investigation of what regularity hypothesis can substitute for the *unavailable* constraint at irrational  $p$ , once it became clear — after Hamel’s 1905 construction — that the rational-coefficient version of the equation does not by itself force affineness. With the continuous-coefficient version, the irrational-coefficient constraint is *available*, and the regularity dictionary no longer has load to bear.

#### 4. VARIANTS AND LIMITS.

**The strict-minimum hypothesis.** The proof of Theorem 1 uses (\*) at exactly one configuration:  $u_1 = M, u_2 = 0$ , with  $p \in [0, 1]$  ranging freely. The full force of (\*) — for every pair  $(u_1, u_2)$  — is not consumed. We record the strict minimum.

**Theorem 3.** Let  $M > 0$  and let  $G: [0, M] \rightarrow \mathbb{R}$  satisfy

$$pG(M) + (1-p)G(0) = G(pM) \quad \text{for all } p \in [0, 1].$$

| Hypothesis on $G$                                                                                                            | For $(J_2) \Rightarrow \text{affine?}$                                                     | For $(\star) \Rightarrow \text{affine?}$              |
|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>Continuity on <math>I</math>:</b> $G$ is continuous at every $v \in I$ .                                                  | Yes, suffices (Cauchy [12]).                                                               | <b>No</b> —it is the <i>conclusion</i> (Corollary 2). |
| <b>Measurability on <math>I</math>:</b> $G$ is Lebesgue- or Borel-measurable on $I$ .                                        | Yes, suffices (Sierpiński [45]).                                                           | <b>No</b> —same.                                      |
| <b>Monotonicity on <math>I</math>:</b> $G$ is non-decreasing or non-increasing on $I$ .                                      | Yes, suffices (Darboux [14]).                                                              | <b>No</b> —same.                                      |
| <b>Boundedness on a set of positive measure:</b> $G$ is bounded on some Lebesgue-measurable $E \subseteq I$ with $ E  > 0$ . | Yes, suffices (Kormes [33]; bounded on one side suffices, Ostrowski [40], Kestelman [30]). | <b>No</b> —same.                                      |
| <b>Boundedness on <math>I</math>:</b> $G$ is bounded on all of $I$ .                                                         | Yes, suffices (Darboux [15]).                                                              | <b>No</b> —same.                                      |
| <b>None:</b> no regularity hypothesis at all.                                                                                | <b>Insufficient</b> —Hamel-basis pathology [25].                                           | <b>Sufficient</b> —Theorem 1.                         |

**Table 1.** Regularity hypotheses on  $G$ . Column 2 records which hypotheses are classically sufficient, one at a time, to force affineness for the discrete-coefficient equation  $(J_2)$ . Column 3 records their status under the continuous-coefficient equation  $(\star)$ . Steinhaus’s difference-set theorem [48] powers the modern proofs of the two measure-theoretic rows but is itself a tool rather than a regularity result; for the historical catalog see Reem [41] and Kuczma [35, §13.2].

Then  $G(v) = G(0) + (G(M) - G(0))v/M$  on  $[0, M]$ .

*Proof.* Set  $p := v/M$ ; then  $p \in [0, 1]$  since  $v \in [0, M]$ . The hypothesis gives

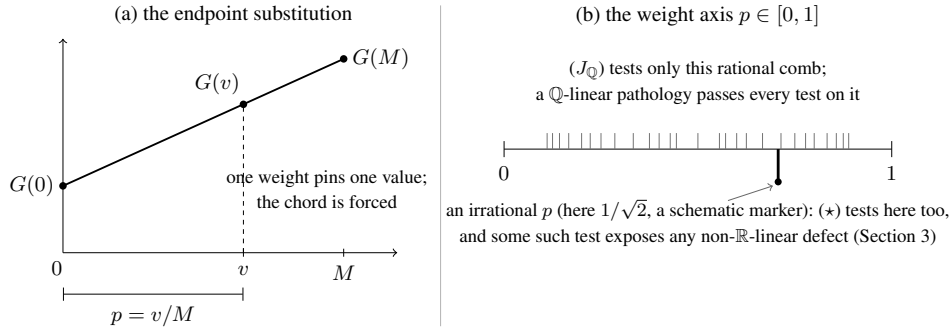
$$pG(M) + (1 - p)G(0) = G(pM) = G(v),$$

so  $G(v) = G(0) + p(G(M) - G(0)) = G(0) + (G(M) - G(0))v/M$ . ■

Theorem 3 is logically weaker than Theorem 1: the one-configuration hypothesis does not a priori imply the full  $(\star)$ . Under the conclusion (affineness) both hold; in practice an author who derives  $(\star)$  from a richer setup (e.g., a Jensen-equality identity in a Bayes-risk computation; see Section 5) typically obtains  $(\star)$  in full force. The narrower Theorem 3 is the right reference when one wants to know exactly how much of the equation the proof actually consumes.

Two further variations sharpen both points: how little input the conclusion needs, and what survives when the full hypothesis is cut down.

**Remark 4 (Irrational weights alone suffice).** The weight set in Theorem 1 need not contain a single rational. Suppose  $(\star)$  holds merely for every *irrational*  $p \in (0, 1)$ , and let  $L(v) := G(0) + (G(M) - G(0))v/M$  denote the affine interpolant of the endpoint values. If  $v/M$  is irrational, the endpoint substitution at  $p = v/M$  gives  $G(v) = L(v)$  directly, and the endpoints  $v \in \{0, M\}$  lie on  $L$  by definition. If  $v/M$



**Figure 2.** The mechanism. (a) The endpoint substitution that proves Theorem 1: at the configuration  $u_1 = M$ ,  $u_2 = 0$ , the single weight  $p = v/M$  pins  $G(v)$  to the chord through  $(0, G(0))$  and  $(M, G(M))$  — one weight per point, the chord forced everywhere. (b) The weight axis on which the two equations live. The rational comb is where  $(J_{\mathbb{Q}})$  tests the identity, and a  $\mathbb{Q}$ -linear Hamel pathology passes every test on the comb. The continuous-coefficient equation  $(\star)$  also tests the irrational weights, where some test must expose any failure of  $\mathbb{R}$ -linearity, and forecloses on the pathology there.

is rational with  $0 < v < M$ , fix the irrational weight  $p = \frac{\sqrt{2}}{2}$  and choose  $\varepsilon > 0$  with  $\varepsilon/M = \frac{\sqrt{3}}{n}$  for an integer  $n$  large enough that  $u_1 := v + (1 - p)\varepsilon$  and  $u_2 := v - p\varepsilon$  lie in  $[0, M]$ . Then  $u_1/M$  and  $u_2/M$  are irrational, being a rational plus the irrational numbers  $\frac{\sqrt{3}}{n} - \frac{\sqrt{6}}{2n}$  and  $-\frac{\sqrt{6}}{2n}$ , respectively (the first is irrational because  $(2\sqrt{3} - \sqrt{6})^2 = 18 - 12\sqrt{2}$  is). Since  $pu_1 + (1 - p)u_2 = v$ , the hypothesis at the triple  $(u_1, u_2, p)$  together with the already-established values  $G(u_i) = L(u_i)$  gives  $G(v) = pL(u_1) + (1 - p)L(u_2) = L(v)$ , because  $L$  is affine. The Hamel pathology is exposed by irrational weights (Section 3); it is fitting that irrational weights alone suffice to exclude it.

**Corollary 5 (Piecewise saturation).** *Let  $0 = m_0 < m_1 < \cdots < m_k = M$  and suppose  $G: [0, M] \rightarrow \mathbb{R}$  satisfies  $(\star)$  separately on each cell  $[m_{i-1}, m_i]$ , that is, for all  $u_1, u_2$  in a common cell and all  $p \in [0, 1]$ . Then  $G$  is affine on each cell, hence piecewise affine on  $[0, M]$  and — being a single function whose affine pieces share their values at the knots — automatically continuous, with no regularity hypothesis. Global affineness does not follow: the slopes may differ across knots.*

*Proof.* Apply Theorem 1 on each cell after the affine change of variable  $v \mapsto (v - m_{i-1})/(m_i - m_{i-1})$ . Continuity at each knot  $m_i$  holds because the affine formulas on the two adjacent cells both evaluate there to the same number,  $G(m_i)$ . ■

The corollary is the form in which Theorem 1 is consumed by the calibration setting of Section 5: a surrogate score on  $[0, 1]$  that aggregates *exactly* on each of  $[0, \frac{1}{2}]$  and  $[\frac{1}{2}, 1]$  is necessarily piecewise affine — a tent-shaped profile once the endpoint and scale normalizations are fixed — with nothing assumed at the kink or anywhere else.

**Convex domains in higher dimensions.** The endpoint substitution extends to any convex subset of a real vector space.

**Theorem 6.** *Let  $V$  be a real vector space (not restricted to finite dimensions), let  $C \subseteq V$  be a nonempty convex set, and let  $G: C \rightarrow \mathbb{R}$  satisfy*

$$pG(x_1) + (1 - p)G(x_2) = G(px_1 + (1 - p)x_2) \quad (x_1, x_2 \in C, p \in [0, 1]).$$



Then there is a linear functional  $a: \text{span}(C - C) \rightarrow \mathbb{R}$  on the parallel subspace  $W := \text{span}(C - C)$  (the subspace parallel to the affine hull of  $C$ ) and a constant  $b \in \mathbb{R}$  such that  $G(x) = a(x - x_0) + b$  on  $C$  for any fixed  $x_0 \in C$ . Equivalently,  $G$  is affine on  $C$ .

*Proof.* Fix  $x_0 \in C$ , let  $D := C - x_0$  (a convex set containing 0), and set  $A(v) := G(x_0 + v) - G(x_0)$  for  $v \in D$ , so that  $A(0) = 0$ .

(i) *Real homogeneity on  $D$ .* For  $v \in D$  and  $p \in [0, 1]$ , the hypothesis at  $x_1 = x_0 + v$ ,  $x_2 = x_0$  gives  $A(pv) = p A(v)$ —the endpoint substitution of Theorem 1, now run along the segment  $[x_0, x_0 + v]$ . This is the step where the continuum of weights acts; everything after it is bookkeeping in which  $\mathbb{Q}$  plays no role.

(ii) *Additivity where defined.* For  $v, w \in D$  the hypothesis at  $x_1 = x_0 + v$ ,  $x_2 = x_0 + w$ ,  $p = \frac{1}{2}$  gives  $A(\frac{v+w}{2}) = \frac{1}{2}(A(v) + A(w))$ ; if moreover  $v + w \in D$ , then (i) at  $p = \frac{1}{2}$  turns the left-hand side into  $\frac{1}{2}A(v + w)$ , whence  $A(v + w) = A(v) + A(w)$  whenever  $v, w, v + w \in D$ .

(iii) *Extension to a linear functional.* On the cone  $D^* := \bigcup_{\lambda > 0} \lambda D$  (which is a convex cone generated by  $D$ ) define  $\hat{A}(\lambda v) := \lambda A(v)$  for  $v \in D$ ,  $\lambda > 0$ . This is well defined: if  $\lambda v = \mu w$  with  $v, w \in D$ , scale by  $t > 0$  small enough that  $t\lambda, t\mu \leq 1$ ; then  $t\lambda v = t\mu w \in D$  (since  $0 \in D$  and  $D$  is convex,  $sD \subseteq D$  for  $s \in [0, 1]$ ), and two applications of (i) give  $\lambda A(v) = \mu A(w)$ . The same scaling device, choosing in the additivity step  $t > 0$  small enough that  $t(\lambda + \mu) \leq 1$ , upgrades (ii) to additivity of  $\hat{A}$  on all of  $D^*$ , and (i) to  $\hat{A}(cu) = c \hat{A}(u)$  for every real  $c > 0$  and  $u \in D^*$ . Finally,  $\text{span}(C - C) = D^* - D^*$ ; for  $u = v - w$  with  $v, w \in D^*$  set  $a(u) := \hat{A}(v) - \hat{A}(w)$ . Additivity of  $\hat{A}$  makes  $a$  well defined and additive, and positive homogeneity extends to full  $\mathbb{R}$ -homogeneity via  $a(-u) = -a(u)$ , so  $a$  is linear. Then  $G(x) = a(x - x_0) + G(x_0)$  on  $C$ , and any function of this form satisfies the hypothesis. ■

The structurally essential step is the one-dimensional endpoint substitution of Theorem 1. The higher-dimensional lifting is largely bookkeeping (compare the regularity-laden treatments of the Jensen equation on convex domains in Kuczma [35, §13.2] and Aczél–Dhombres [3]).

**The rational-coefficient version retains the pathology.** The strengthening from rational to continuous coefficients in  $(\star)$  does genuine work: the rational-coefficient version  $(J_{\mathbb{Q}})$  retains the Hamel pathology.

**Proposition 7 (Folklore; cf. [35, §§5.2 and 13.2]).** *There exists  $G: [0, 1] \rightarrow \mathbb{R}$  satisfying  $(J_{\mathbb{Q}})$  (hence  $(J_2)$ ) on  $[0, 1]$  that is not affine.*

*Construction.* Choose a Hamel basis  $H_0$  of  $\mathbb{R}$  over  $\mathbb{Q}$  containing 1; such a basis exists by Zorn’s lemma, and some choice is genuinely needed: in Solovay’s model of  $\text{ZF} + \text{DC}$  every set of reals is Lebesgue measurable [47], so there every solution of the Cauchy or Jensen equations is affine. Pick any  $h' \in H_0 \setminus \{1\}$ ; since two distinct rationals are  $\mathbb{Q}$ -linearly dependent,  $h'$  is irrational. Choose a nonzero rational  $q$  with  $h := qh' \in (0, 1)$ —possible because the rational multiples of  $h'$  are dense in  $\mathbb{R}$ —and set  $H := (H_0 \setminus \{h'\}) \cup \{h\}$ . Replacing a basis element by a nonzero rational multiple of itself preserves both  $\mathbb{Q}$ -linear independence and spanning, so  $H$  is again a Hamel basis, and it contains 1 together with the irrational  $h \in (0, 1)$ .<sup>1</sup> Define a  $\mathbb{Q}$ -linear map

<sup>1</sup>Cardinality alone does not guarantee a basis element lies in  $(0, 1)$ : rescaling every element of a Hamel basis into  $[1, 2]$ , say, produces a basis that avoids  $(0, 1)$  entirely. It is easy to overlook this domain point; the rational-rescaling device used here is standard (Reem [41, Remark 3.4] uses it, after an idea of Z. Boros, to produce a Hamel basis dense in  $\mathbb{R}$ ) and guarantees  $h \in (0, 1)$  while keeping basis invariants.

$\ell: \mathbb{R} \rightarrow \mathbb{R}$  by setting  $\ell(1) = 0$ ,  $\ell(h) = 1$ , and extending  $\mathbb{Q}$ -linearly on the remaining basis elements.

By construction,  $\ell$  is  $\mathbb{Q}$ -linear:  $\ell(x + y) = \ell(x) + \ell(y)$  (Cauchy's equation) and  $\ell(qx) = q\ell(x)$  for all  $q \in \mathbb{Q}$  ( $\mathbb{Q}$ -homogeneity). Set  $G := \ell|_{[0,1]}$ ; then  $G$  satisfies  $(J_{\mathbb{Q}})$  on  $[0, 1]$  because  $\mathbb{Q}$ -rational convex combinations of points of  $[0, 1]$  remain in  $[0, 1]$  and  $\ell$  respects  $\mathbb{Q}$ -linear combinations.

Explicit witness that  $G$  is not affine:  $G(h) = \ell(h) = 1 \neq 0$ , while  $G(q) = \ell(q) = q\ell(1) = 0$  for every rational  $q \in [0, 1]$ . This  $G$  oscillates wildly: its graph is dense in  $[0, 1] \times \mathbb{R}$  ( $G$  takes the value  $r$  on the dense set  $(q + rh)_{q \in \mathbb{Q}} \cap [0, 1]$ , for every rational  $r$ ), so continuity, monotonicity, and boundedness fail spectacularly; and  $G$  can satisfy *none* of the hypotheses of Table 1, since by that table's second column any one of them would force affineness. Any affine map  $A: [0, 1] \rightarrow \mathbb{R}$  agreeing with  $G$  on  $\mathbb{Q} \cap [0, 1]$  would satisfy  $A(0) = 0$  and  $A(q) = 0$  for all rational  $q$ —forcing  $A \equiv 0$ , contradicting  $G(h) = 1$ . ■

Proposition 7 makes sharp the dictionary:  $(J_{\mathbb{Q}})$  on  $[0, 1]$  admits non-affine solutions; only the strengthening to the continuous-coefficient  $(\star)$  retires the regularity-hypothesis requirement.

**5. TELLING  $(\star)$  FROM ITS RELATIVES: A FIELD GUIDE TO SATURATED JENSEN.** Theorem 1 closes the continuous-coefficient equation  $(\star)$  with no regularity hypothesis on  $G$ ; Proposition 7 shows that the same closure fails for the rational-coefficient relative  $(J_{\mathbb{Q}})$ , where the Hamel pathology survives. A derivation in an applied field that pushes Jensen's inequality to equality therefore lands on one side or the other of a sharp line, and which side it lands on decides whether a regularity hypothesis is doing real work. The purpose of this section is to draw that line and show how to use it. The conclusion is not that applied authors habitually carry unnecessary hypotheses; it is more precise and, we think, more useful: the saturated-Jensen identity arises from at least four structurally different sources, only one of which is the regularity-free situation of Theorem 1, and the four are easily mistaken for one another.

**Four sources of a saturated identity, and the questions that separate them.** A saturated Jensen identity

$$pG(u_1) + (1 - p)G(u_2) = G(pu_1 + (1 - p)u_2)$$

is, first of all, a *family* of assertions indexed by the weight  $p$ . Its mathematical content is precisely the set of weights for which the surrounding derivation actually licenses the assertion. Four recurring sources are worth separating.

The first is **finite mixing**. If the only primitive in the model is a binary mixture—combine two objects in some proportion—and compound objects are built by finitely iterating it, then the weights one can actually reach are the dyadic rationals, and, after closure under the operation, at most the rationals in  $[0, 1]$ . Such a derivation licenses  $(J_{\mathbb{Q}})$ , not  $(\star)$ ; by Proposition 7 a regularity hypothesis is then genuinely required, and Theorem 1 does not apply.

The second is **additivity into a bounded codomain**. If the derivation delivers additivity of  $G$  together with the information that  $G$  takes values in a bounded set, then  $G$  is affine by the classical theorem that a bounded additive function is linear—the very results recorded on the two boundedness struts of Figure 1. Here a regularity hypothesis (boundedness) is load-bearing, but it is supplied for free by the codomain rather than assumed by hand, and again the operative mechanism is not the endpoint substitution.

The third is a **Cauchy relative with a non-affine target**. If the equation that actually appears is a relative of Cauchy’s equation whose intended solution is *not* affine, then a regularity hypothesis is needed to exclude Hamel-type pathological solutions, and  $(\star)$  is simply not the equation in play. The characterization of Shannon entropy is the textbook instance (treated below).

The fourth is a **genuine continuum of real weights**. If the weight  $p$  is a real parameter realized inside the structure — an atomless measure assigns to events every mass in  $[0, 1]$ , by Sierpiński’s intermediate-value theorem for atomless measures [46], of which Lyapunov’s convexity theorem [37] is the vector-valued descendant — and the saturated identity is asserted *directly* for that real  $p$ , with no detour through finite iteration, then the derivation licenses  $(\star)$  at every real  $p$ , irrationals included. Only now does Theorem 1 apply, and only now is a regularity hypothesis on  $G$  vestigial.

This yields a simple ordered diagnostic. The *zeroth question* — the name we use throughout, and a gate one must clear before any weight is examined — is logically prior to the four sources: *does a real-valued solution of the identity provably exist?* Theorem 1 takes a function as input and reports its form; it cannot conjure the function, and an axiom system whose representation theorem fails (the expected-utility example below) fails here, before any weight is examined. Once this gate is cleared, the question that separates the four sources is: *does the derivation assert the identity at every real weight because the weights form a genuine continuum in the structure — or does it merely reach the rational weights, lean on the boundedness of the codomain, or solve a Cauchy relative whose answer is not affine?* In the continuum case the regularity hypotheses of Figure 1 are vestigial and should be dropped; in the others they are load-bearing and must be kept. We do not claim the taxonomy is exhaustive — countable ( $\sigma$ -additive) mixtures, for instance, assert a strictly stronger identity with a theory of its own — but the four sources cover the derivations we have encountered.

The trap, then, is **misclassification**, and it runs both ways. A derivation of one of the first three kinds can be mistaken for the fourth, so that an author drops a hypothesis that was in fact indispensable. Conversely — and this is the rarer and more seductive error — a derivation of the fourth kind can be mistaken for a Cauchy problem, so that an author *adds* a defensive hypothesis that was never needed. The present paper exists because of an instance of the second error; we return to it in the closing note on provenance (Section 6).

**Where the hypotheses are load-bearing.** We begin on the side of the line where the classical hypotheses do real work — which, perhaps against first intuition, is where most of the familiar “saturated Jensen” derivations live.

**Expected-utility representation (a continuum that is only apparent).** In the von Neumann–Morgenstern tradition [52], sharpened by Herstein and Milnor [26], a preference relation over lotteries that is complete, transitive, independent, and continuous in the Archimedean (mixture-continuity) sense is represented by a utility functional that is linear in probability,

$$U(p L_1 + (1 - p) L_2) = p U(L_1) + (1 - p) U(L_2),$$

and these conditions are jointly necessary and sufficient (see also [21, 34]). It is tempting to read the displayed identity as  $(\star)$  and to offer Theorem 1 as a continuity-free substitute for the Archimedean axiom. That reading is too quick. The independence axiom is a condition on the *preference relation*; although it is stated for every real mixing weight, it does not by itself deliver a cardinal functional equation for a real-valued  $U$ .

What produces a real-valued affine  $U$  in the first place is the **Archimedean axiom** (mathematically, the mixture-continuity condition that embeds the indifference classes into the real line), and it is genuinely indispensable: lexicographic preferences are complete, transitive, and independent, yet violate the Archimedean axiom and admit no real-valued representation whatsoever — so there is no  $U$ , and the displayed identity, and with it  $(\star)$ , is vacuous. Theorem 1 cannot replace the Archimedean axiom because Theorem 1 *presupposes* a real-valued solution of  $(\star)$ , and the Archimedean axiom is exactly what is needed to produce one. In the diagnostic’s terms, this failure is not about a shortage of weights — independence is stated at every real  $p$  — but a failure at the zeroth question: there is no real-valued unknown for the equation to constrain. The Archimedean axiom is load-bearing one step *before* the dictionary is even consulted.

**The axiomatic characterization of Shannon entropy (a Cauchy relative).** In the tradition of Khinchin [31] and Faddeev [20], the entropy  $H(p_1, \dots, p_n) = -\sum_i p_i \log p_i$  is singled out by symmetry, a normalization, and a recursivity (grouping) axiom. The recursivity axiom does not reduce to  $(\star)$ . It reduces to the **fundamental equation of information**, evaluated on the domain  $D = \{(x, y) \in [0, 1]^2 \mid x + y \leq 1\}$ ,

$$f(x) + (1 - x) f\left(\frac{y}{1 - x}\right) = f(y) + (1 - y) f\left(\frac{x}{1 - y}\right),$$

for the information function  $f(x) = H(x, 1 - x)$ , and the unique solution under the axioms (with Faddeev’s normalization  $f(\frac{1}{2}) = 1$ ) is the binary entropy  $f(x) = -x \log_2 x - (1 - x) \log_2 (1 - x)$ , which is strictly *concave* — emphatically not affine. The fundamental equation of information is a relative of Cauchy’s equation. In its *binary* ( $n = 2$ ) form it admits Hamel-type non-measurable solutions unless a regularity hypothesis intervenes [2], and the hypotheses used to exclude them belong to the same regularity families represented by the five struts of Figure 1. Faddeev assumed continuity of  $f$  outright [20] (a clean modern statement is in [6]); the requirement was then weakened one hypothesis at a time: to integrability (Tverberg [51]), measurability (Lee [36]), boundedness on a set of positive measure (Diderrich [17]), and monotonicity on  $[0, \frac{1}{2})$  (Kendall [29], Borges [9]). The entropy characterization is thus the canonical load-bearing case in the dictionary of Section 3: the  $n = 2$  binary equation is a Cauchy relative at rational mixing weights with a non-affine target, and every strut in Figure 1(a) is genuinely needed.

There is an instructive complement, recorded in Aczél and Daróczy [2]: Fischer (1972) proved that for any *single* fixed  $n > 2$ , the same family of grouping equations characterizes the Shannon entropy *without any regularity hypothesis*. The reason is the same as Theorem 1: with  $n > 2$  outcomes, the grouping axiom must hold for all probability vectors — including those with irrational components — so the equation tests  $f$  at a continuum of weights and the Hamel pathology has nowhere to live. The load-bearing vs. vestigial asymmetry thus runs exactly through  $n = 2$  vs.  $n > 2$ , and is another instance of the mechanism this paper diagnoses. We record the  $n = 2$  case as the load-bearing example in the mirror table below.

**Linear opinion pools (boundedness from the codomain).** A third mechanism is worth naming briefly, because it too is readily mistaken for  $(\star)$ . If several experts’ probability assignments are aggregated by a rule that commutes with marginalization, the rule must be a weighted average — a linear opinion pool — as soon as the underlying space has at least three points (McConway [38]; Aczél and Wagner [4]). The aggregation function is additive across disjoint events, and it takes values in  $[0, 1]$  because its outputs are probabilities; linearity then follows from the bounded-additive theorem of Figure 1, not

from the endpoint substitution. As in the utility case, the saturated identity is real, but the work is done by a classical regularity property — here boundedness, supplied by the codomain — rather than by a continuum of weights.

**Where the hypotheses are vestigial.** Now we cross the line, where the continuum of weights is real and Theorem 1 does its work.

**Surrogate calibration on the resolution axis.** We first fix the vocabulary, because the central object here travels under many names. Let  $(\Omega, \mathcal{F}, \mu)$  be an atomless probability space, let  $Y \in \{0, 1\}$  be a binary outcome, and let  $\mathcal{F}_0 \subseteq \mathcal{F}$  be a sub- $\sigma$ -algebra of features (or covariates). The posterior probability profile is the  $\mathcal{F}_0$ -measurable random variable  $\eta := \mathbb{E}[Y \mid \mathcal{F}_0] = \Pr(Y = 1 \mid \mathcal{F}_0) \in [0, 1]$ . For an action space  $\mathcal{A}$  and a loss function  $\ell: \mathcal{A} \times \{0, 1\} \rightarrow \mathbb{R}$ , the *conditional Bayes risk* at posterior value  $u \in [0, 1]$  is  $H(u) := \inf_{a \in \mathcal{A}} ((1 - u)\ell(a, 0) + u\ell(a, 1))$ ; the function  $H$  is variously called the Bayes envelope, the generalized entropy, or the uncertainty function of the associated score [43, 16, 24, 23, 42]. For the 0–1 loss and binary actions  $\mathcal{A} = \{0, 1\}$ , it is the symmetric tent function  $H(u) = \min(u, 1 - u)$ . We write  $G: [0, 1] \rightarrow \mathbb{R}$  for a candidate uncertainty function and, for now, let it be *arbitrary*.

A sub- $\sigma$ -algebra  $\mathcal{G} \subseteq \mathcal{F}_0$  is a *resolution*: the coarser  $\mathcal{G}$ , the less the forecast distinguishes states of the world. Writing  $\eta_{\mathcal{G}} := \mathbb{E}[\eta \mid \mathcal{G}] = \mathbb{E}[Y \mid \mathcal{G}]$ , the risk attainable at resolution  $\mathcal{G}$  is

$$U_G(\mathcal{G}) := \mathbb{E}[G(\eta_{\mathcal{G}})], \quad \text{so that} \quad U_G(\sigma(A)) = \mu(A)G(u_1) + \mu(A^c)G(u_2)$$

when  $\eta$  takes the value  $u_1$  on an event  $A \in \mathcal{F}_0$  and  $u_2$  on its complement  $A^c$ . The classical theory runs along this axis with concavity in hand. Every Bayes envelope is concave (an infimum of affine functions of  $u$ ), and for concave  $G$  the functional  $U_G$  decreases under refinement — conditional Jensen plus the tower property — which is Blackwell’s comparison of experiments [8] in this setting and the refinement ordering of DeGroot and Fienberg [16]. The drop it measures is the *resolution* (or discrimination) term in the decomposition of a proper score: Murphy’s classical partition of the Brier score [39], Bröcker’s generalization to every proper score via its entropy and divergence [10], and the empirical form  $\bar{S} = \text{MCB} - \text{DSC} + \text{UNC}$  of Dimitriadis, Gneiting, and Jordan [18]. Every statement in this classical chain *assumes* concavity of  $G$ , and the empirical theory adds measurability or smoothness besides.

The dictionary’s contribution is the converse direction, and it needs none of those assumptions.

**Proposition 8 (Resolution-blind uncertainty functions are affine).** *Let  $(\Omega, \mathcal{F}, \mu)$  be atomless and let  $G: [0, 1] \rightarrow \mathbb{R}$  be arbitrary. Suppose the two-cell resolutions are exact: for all  $u_1, u_2 \in [0, 1]$  and every event  $A$ ,*

$$\mu(A)G(u_1) + \mu(A^c)G(u_2) = G(\mathbb{E}[\eta]), \quad \text{where } \eta = u_1 \mathbf{1}_A + u_2 \mathbf{1}_{A^c}.$$

*Then  $G$  is affine; consequently  $U_G(\mathcal{G}) = G(\mathbb{E}[\eta])$  for every  $[0, 1]$ -valued posterior  $\eta$  and every sub- $\sigma$ -algebra  $\mathcal{G}$ , so  $G$  is blind to resolution at all scales. No continuity, measurability, monotonicity, boundedness — or concavity — is assumed. The  $k$ -class case, with  $\eta$  valued in the probability simplex  $\Delta_k$  and  $G$  arbitrary on  $\Delta_k$ , holds verbatim with “affine on the simplex,” by Theorem 6.*

*Proof.* Atomlessness realizes every mass  $p = \mu(A) \in [0, 1]$  (Sierpiński [46]), and the values  $u_1, u_2$  are prescribed independently of  $A$ , so the hypothesis asserts  $pG(u_1) +$

$(1 - p) G(u_2) = G(p u_1 + (1 - p) u_2)$  at every triple — the identity  $(\star)$  on its entire domain, with  $G$  only ever *evaluated* at points, never integrated. Theorem 1 gives affineness on  $[0, 1]$ ; Theorem 6 with  $C = \Delta_k$  gives the simplex case. Conversely, for affine  $G$  and any  $\mathcal{G}$ , linearity of expectation and the tower property give  $\mathbb{E}[G(\eta_{\mathcal{G}})] = G(\mathbb{E}[\eta_{\mathcal{G}}]) = G(\mathbb{E}[\eta])$ . ■

A concrete instance is already in the published record, and it shows the dictionary separating clauses that the original framework states together. Reid and Williamson [42] build their integral representation of proper losses on a condition they call *fairness*: the partial losses  $\eta \mapsto \ell(0, \eta)$  and  $\eta \mapsto \ell(1, \eta)$  are respectively right-continuous at 0 and left-continuous at 1, and  $\ell(0, 0) = \ell(1, 1) = 0$ . Stated together, the two clauses read as one regularity hypothesis on the loss; the dictionary of Section 3 pulls them apart. The boundary-continuity clause is precisely a strut of Figure 1, and on an atomless feature space it is *vestigial*: by Proposition 8 any uncertainty function arising from cellwise-exact aggregation is affine on each cell, hence continuous at the endpoints, so the continuity is a *conclusion* rather than an input. The zero-loss clause is not of this kind at all. Theorem 1 determines the uncertainty function only up to its two endpoint values  $G(0)$  and  $G(M)$ ; the condition  $\ell(0, 0) = \ell(1, 1) = 0$  is exactly what pins those two free constants, and no continuum of weights can supply it. In those terms it lies past the zeroth question but is a *normalization*, not a Hamel-excluding regularity hypothesis: dropping it does not readmit the pathology, it merely unfixes an additive and a slope constant. The lesson is the mirrored dictionary in miniature — one clause of a single published hypothesis is vestigial under  $(\star)$ , the other is load-bearing for a reason that has nothing to do with  $(\star)$  — and conflating the two is exactly the misreading the diagnostic above is built to prevent.

This is the tempting misclassification warned of above in miniature: the saturated identity looks like a Cauchy problem and invites a defensive boundedness or measurability hypothesis on  $G$ , but the continuum of cell masses makes it genuine  $(\star)$  and the hypothesis vestigial.

**The benchmark partition and the tent.** What survives when exactness holds only *locally* is Corollary 5: a surrogate that aggregates exactly on each cell of a fixed benchmark partition of the probability axis is affine on each cell, with continuity at the knots for free. For the canonical benchmark  $\{[0, \frac{1}{2}], [\frac{1}{2}, 1]\}$  the resulting shape is piecewise affine; with the usual 0–1 endpoint and scale normalizations, this becomes the familiar symmetric tent, whose probabilistic content is fixed by a classical computation. For the cost-weighted loss  $\ell_c$  — the cheapest misclassification primitive, which charges  $c$  for a false positive and  $1 - c$  for a false negative — Reid and Williamson [42, Thm 14] compute the conditional Bayes risk explicitly as the tent

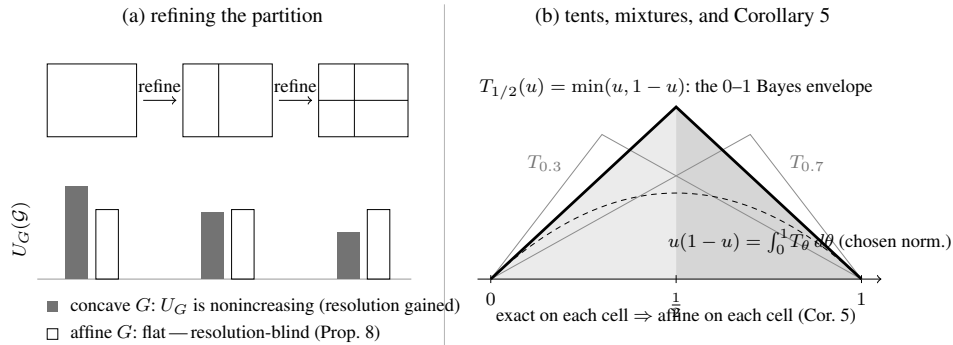
$$L_c(u) = \min\{(1 - u)c, (1 - c)u\},$$

a function concave in both arguments, with apex at  $u = c$  and height  $c(1 - c)$ . Setting  $c = \frac{1}{2}$  and recalling  $2\ell_{1/2} = \ell^{0-1}$  returns the symmetric envelope  $\min(u, 1 - u)$ ; the apexed tents  $T_\theta$  of the elementary scores [44, 19] are these same  $L_\theta$  up to the 0–1 normalization. What our Corollary 5 adds is the *converse*: piecewise saturation on the benchmark cells pins the piecewise-affine shape; the endpoint and scale normalizations then select the tent, with affineness on each cell and continuity at the apex obtained for free and no regularity assumed. The mixture representation is equally concrete. Under the usual binary proper-score representation, every suitably normalized proper score’s uncertainty function  $G$  is recovered from the tents by integrating against the curvature

of the Bayes risk itself—Reid and Williamson’s weight function [42, Thm 16] is  $w(c) = -L''(c) \geq 0$ , so

$$G = \int_0^1 T_\theta d\nu(\theta), \quad d\nu(\theta) = -G''(\theta) d\theta,$$

identifying the abstract mixing measure  $d\nu$  on the extreme rays with the (distributional) second-derivative measure of  $G$ . The uniform mixture  $\int_0^1 T_\theta d\theta = u(1-u)$  with this normalization is then the Brier entropy, the case  $w \equiv \text{const}$ . The tents are the extreme rays of the cone of uncertainty functions; Theorem 1 explains why exactness selects them, and [42, Thms 14, 16] say what they are. In learning-theoretic language, cellwise exactness is tightness of the surrogate-risk bracket: the  $\psi$ -transform of Bartlett, Jordan, and McAuliffe [7], Steinwart’s calibration functions [49], and the multiclass analysis of Tewari and Bartlett [50] measure how far a surrogate’s floor sits from the true floor; exactness on a cell says the gap is zero there, and Corollary 5 says the price of zero gap is affineness on the cell. Figure 3 draws both halves.



**Figure 3.** Saturation along the resolution axis. (a) Refining the partition can only lower the uncertainty functional  $U_G(\mathcal{G}) = \mathbb{E}[G(\eta_{\mathcal{G}})]$  when  $G$  is concave (conditional Jensen; Blackwell [8]); the drop is the resolution term of the score decompositions [39, 10, 18]. By Proposition 8, the only uncertainty functions whose profile is flat—resolution-blind at every two-cell comparison—are the affine ones, with no regularity assumed. (b) The tent  $T_{1/2}(u) = \min(u, 1-u)$  is the 0–1 Bayes envelope; the tents  $T_\theta$  are the uncertainty functions of the elementary scores [44, 19], and the usual binary proper-score representation writes suitably normalized uncertainty functions as mixtures  $\int_0^1 T_\theta d\nu(\theta)$ ; the dashed curve is the Brier entropy  $u(1-u)$ , the uniform mixture under the chosen tent normalization. Exactness on each cell of the benchmark partition  $\{[0, \frac{1}{2}], [\frac{1}{2}, 1]\}$  forces affineness on each cell (Corollary 5): a piecewise-affine tent shape, with the familiar symmetric tent selected by endpoint and scale normalizations, pinned with nothing assumed at the kink.

**Two readings.** The same reading travels. In probabilistic graphical models, a clique tree is *calibrated* when neighboring cliques agree on their sepset marginals [32]; this cellwise agreement is the piecewise identity above, asserted over the sepset partition, and  $U_G$  is the natural uncertainty functional along the junction tree’s resolution order. In representation learning, an encoder  $R$  induces the resolution  $\mathcal{G} = \sigma(R)$ , and  $U_{T_{1/2}}(\sigma(R))$  is the smallest misclassification error achievable from the representation—its error floor. Proposition 8 then says that the only uncertainty functions blind to the choice of representation are the affine ones: a genuinely concave  $G$  can certify a resolution gain through a strict drop of  $U_G$ , provided the refinement crosses a segment on which  $G$  is strictly concave.

**The thin line, in miniature.** Replace the atomless space by a single fair coin — build the mixtures by tossing it repeatedly — and the attainable masses  $p$  collapse to the dyadic rationals; one is back in the first source of Section 5, Proposition 7 returns a non-affine solution, and a regularity hypothesis is once more required.<sup>2</sup> The main contrast of Sections 3 and 5 is visible in the gap between an atomless measure and a coin.

**Mixture-affine assignments on quantum states.** A second instance of the fourth source — this one consuming the higher-dimensional Theorem 6 — arises in the foundations of quantum mechanics. The density operators  $\mathcal{S}_d$  on  $\mathbb{C}^d$  form a convex set, and the statistical mixture  $p\rho_1 + (1-p)\rho_2$  is operationally primitive: prepare  $\rho_1$  with probability  $p$  using a classical randomizer, with  $p$  ranging over a genuine continuum. Let  $v: \mathcal{S}_d \rightarrow \mathbb{R}$  be any assignment respecting mixtures,

$$v(p\rho_1 + (1-p)\rho_2) = pv(\rho_1) + (1-p)v(\rho_2) \quad (\rho_1, \rho_2 \in \mathcal{S}_d, p \in [0, 1]).$$

This is exactly the hypothesis of Theorem 6 on the convex set  $\mathcal{S}_d$ , so  $v$  extends to a linear-plus-constant functional; since every linear functional on the finite-dimensional real space of Hermitian operators has the form  $u \mapsto \text{tr}(Bu)$ , we get  $v(\rho) = \text{tr}(A\rho) + b$  for a Hermitian  $A$  — with no positivity, boundedness, continuity, or measurability assumed of  $v$ . The affine representation itself is operational folklore (see, e.g., Holevo [27, Lemma 1.6.1]); what the dictionary adds is a diagnosis of why the dual derivations must pay more. Gleason-type theorems work on the *measurement* side: a frame function on effects is *finitely* additive over coarse-grainings of measurements (where grouping disjoint measurement outcomes corresponds to operator addition,  $E_1 + E_2 \leq I$ ), and coarse-graining is a finite-additivity operation rather than a direct continuum of real convex mixtures — the first source. Finite additivity yields homogeneity only over the nonnegative rationals, so the rational-to-real step must be purchased with a regularity hypothesis. Each classical derivation pays in its own coin. Gleason’s original projection-based theorem [22] uses nonnegativity (for  $d \geq 3$ ), as does Busch’s POVM version [11] (for all  $d \geq 2$ ); Caves, Fuchs, Manne, and Renes [13] use positivity or an explicit continuity argument. The Cauchy-equation treatment of Wright and Weigert [53] (Theorem 1, with a corrected proof in [54]) makes the dependence sharpest: a nonlinear finitely additive frame function built on a Hamel basis can be neither bounded on either side, nor continuous at zero, nor Lebesgue measurable — the catalog of Figure 1 rediscovered inside physics. State preparation tests the identity directly at every real weight  $p \in [0, 1]$  via the classical randomizer; measurement coarse-graining tests it only through additivity, leaving room for dense Hamel-basis oscillations unless boundedness or positivity intervenes. The dictionary predicts which route needs a rational-to-real bridge, and the derivations cited above supply such a bridge through positivity, boundedness, or continuity hypotheses.

**The dictionary, mirrored.** Figure 1 records, for a *single* equation, which hypotheses force affineness. The diagnostic of this section is that the *same five hypotheses* sit on opposite sides depending on the source of the identity. Table 2 places the load-bearing case above (the entropy characterization, a Cauchy relative) beside the vestigial case (the atomless calibration computation, genuine  $(\star)$ ); the contrast between its two columns is the content of the section.

<sup>2</sup>A terminological caution: the property tested throughout is that  $G$  commutes with expectation over two-point laws — we say the family is *expectation-exact* — rather than “mean-preserving,” a term that already names mean-preserving spreads in economics.



| Hypothesis on the unknown function       | In the entropy characterization (a Cauchy relative)                                                                                                                   | Under a saturated ( $\star$ ) on an atomless space |
|------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Continuity                               | Load-bearing (Faddeev [20]).                                                                                                                                          | Vestigial — a conclusion (Corollary 2).            |
| Measurability                            | Load-bearing (Lee [36]).                                                                                                                                              | Vestigial — same.                                  |
| Monotonicity                             | Load-bearing (Kendall [29], Borges [9]).                                                                                                                              | Vestigial — same.                                  |
| Boundedness on a set of positive measure | Load-bearing (Diderrich [17]).                                                                                                                                        | Vestigial — same.                                  |
| Boundedness on the interval              | Load-bearing (a fortiori).                                                                                                                                            | Vestigial — same.                                  |
| None                                     | Insufficient — Hamel-type solutions of the <i>binary</i> ( $n=2$ ) fundamental equation of information [2] (for $n>2$ , Fischer 1972 showed no such solutions exist). | Sufficient — Theorem 1.                            |

**Table 2.** The dictionary of Figure 1, mirrored. The same five hypotheses serve as load-bearing regularity routes for the *binary* ( $n=2$ ) fundamental equation of information (Cauchy relative at rational weights; unique regular solution is the concave Shannon entropy), and vestigial when a continuum of weights makes the operative equation the continuous-coefficient ( $\star$ ). For  $n>2$ , Fischer (1972) showed no regularity is needed (see §5).

**The lesson.** The recurrence worth articulating is less a habit of carrying unnecessary hypotheses than the ease of misreading which of the four sources one is in. Saturated Jensen is produced by finite mixing, by additivity into a bounded codomain, by Cauchy relatives with non-affine targets, and by genuine continua of real weights, and only the last retires the regularity hypotheses. The zeroth question precedes it: absent an existence axiom there is no real-valued unknown, and the question of weights never arises. The diagnostic above is offered as a way to tell them apart; the mirrored dictionary of Section 5 is the warning that the same five classical hypotheses appear on both sides. An author who has internalized the Cauchy–Hamel toolkit is well defended against the first error and peculiarly exposed to the second — which is, as the closing note on provenance recounts, how this paper began.

**6. CONCLUDING REMARKS.** Theorem 1 is, mathematically, one line. The dictionary of Section 3 is a table. The structural argument of Section 3 is one sentence: *the asymmetry above holds; the continuous-coefficient equation ( $\star$ ) forecloses on it there.* Why does this material deserve an article in the MONTHLY as an expository matter?

Two reasons. First, the classical detour through Hamel and the Polish school produced *foundational* results of classical analysis — and the dictionary of Section 3 articulates a precise sense in which the detour answered an adjacent question: not

“what is the minimum hypothesis for  $(\star) \Rightarrow \text{affine}$ ?” (Theorem 1’s answer: nothing), but “what is the minimum hypothesis for  $(J_2) \Rightarrow \text{affine}$ ?” (Polish school’s answer: any of a dictionary of tameness conditions). Both questions are interesting; their answers are different.

Second, the diagnostic of Section 5 is, to our knowledge, not articulated explicitly elsewhere in the functional-equations literature. The same five regularity hypotheses turn out to be indispensable in some saturated-Jensen settings (the Shannon-entropy characterization) and vestigial in others (a calibration computation on an atomless space), and the explicit criterion may save other authors from misreading which case they are in — in either direction. Nor is the asymmetry hypothetical: in the Gleason-type derivations of the quantum Born rule, the measurement-side route must purchase the rational-to-real step with positivity or one of the dictionary’s hypotheses [22, 11, 13, 53, 54], while the preparation-side route of Section 5 pays nothing — the two columns of Table 2, alive in a single literature.

The endpoint substitution itself is the kind of one-line proof that, once seen, is hard to unsee — but that nonetheless takes some scaffolding to make visible. That scaffolding is the paper’s expository purpose.

**A note on provenance.** The present observation reached the author through an unconventional route: a formalization project on a separate topic, in which the lemma “ $G$  satisfies  $(\star)$  on  $[0, M]$  implies  $G$  is affine” was initially stated with a defensive boundedness hypothesis on  $G$ , in deference to the Cauchy–Hamel literature. When the formal proof was written out, the boundedness hypothesis turned out to be unused: the endpoint substitution closed the proof on its own. A literature search then placed the substitution technique in its folklore context: Aczél’s [1, §2.1.4] proves a dyadic precursor under a continuity hypothesis. The precise continuous-coefficient regularity-free statement of Theorem 1, while elementary, does not appear in this exact form in the standard references the author consulted; the present paper grew out of the attempt to explain, to the author’s own satisfaction, why the boundedness hypothesis had felt necessary in the first place, and to articulate the contrast between the discrete- and continuous-coefficient cases as a dictionary. The lesson is broader: a defensive regularity hypothesis is not free, and a formalization project occasionally surfaces such hypotheses precisely because the formal system requires every clause of the statement to be used.

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## REFERENCES

- [1] Aczél, J. (1966). *Lectures on Functional Equations and Their Applications*. New York: Academic Press.

- [2] Aczél, J., Daróczy, Z. (1975). *On Measures of Information and Their Characterizations*. New York: Academic Press.
- [3] Aczél, J., Dhombres, J. (1989). *Functional Equations in Several Variables*. Cambridge: Cambridge Univ. Press.
- [4] Aczél, J., Wagner, C. (1980). A characterization of weighted arithmetic means. *SIAM J. Algebraic Discrete Methods*. 1(3): 259–260.
- [5] Aumann, R. J. (1964). Markets with a continuum of traders. *Econometrica*. 32(1–2): 39–50.
- [6] Baez, J. C., Fritz, T., Leinster, T. (2011). A characterization of entropy in terms of information loss. *Entropy*. 13(11): 1945–1957. doi.org/10.3390/e13111945
- [7] Bartlett, P. L., Jordan, M. I., McAuliffe, J. D. (2006). Convexity, classification, and risk bounds. *J. Amer. Statist. Assoc.* 101(473): 138–156. doi.org/10.1198/016214505000000907
- [8] Blackwell, D. (1953). Equivalent comparisons of experiments. *Ann. Math. Statist.* 24: 265–272.
- [9] Borges, R. (1967). Zur Herleitung der Shannonschen Information. *Math. Z.* 96: 282–287.
- [10] Bröcker, J. (2009). Reliability, sufficiency, and the decomposition of proper scores. *Q. J. R. Meteorol. Soc.* 135(643): 1512–1519. doi.org/10.1002/qj.456
- [11] Busch, P. (2003). Quantum states and generalized observables: a simple proof of Gleason’s theorem. *Phys. Rev. Lett.* 91: 120403. doi.org/10.1103/PhysRevLett.91.120403
- [12] Cauchy, A. L. (1821). *Cours d’analyse de l’École royale polytechnique. Première partie: Analyse algébrique*. Paris: Imprimerie royale.
- [13] Caves, C. M., Fuchs, C. A., Manne, K. K., Renes, J. M. (2004). Gleason-type derivations of the quantum probability rule for generalized measurements. *Found. Phys.* 34(2): 193–209. doi.org/10.1023/B:FOOP.0000019581.00318.a5
- [14] Darboux, G. (1875). Sur la composition des forces en statique. *Bull. Sci. Math. Astron.* 9: 281–288.
- [15] Darboux, G. (1880). Sur le théorème fondamental de la géométrie projective. *Math. Ann.* 17: 55–61.
- [16] DeGroot, M. H., Fienberg, S. E. (1983). The comparison and evaluation of forecasters. *J. Roy. Statist. Soc. Ser. D (The Statistician)*. 32(1–2): 12–22.
- [17] Diderich, G. T. (1975). The role of boundedness in characterizing Shannon entropy. *Inform. and Control*. 29(2): 149–161.
- [18] Dimitriadis, T., Gneiting, T., Jordan, A. I. (2021). Stable reliability diagrams for probabilistic classifiers. *Proc. Natl. Acad. Sci. USA*. 118(8): e2016191118. doi.org/10.1073/pnas.2016191118
- [19] Ehm, W., Gneiting, T., Jordan, A., Krüger, F. (2016). Of quantiles and expectiles: consistent scoring functions, Choquet representations, and forecast rankings. *J. R. Stat. Soc. Ser. B.* 78(3): 505–562. doi.org/10.1111/rssb.12154
- [20] Faddeev, D. K. (1957). Zum Begriff der Entropie eines endlichen Wahrscheinlichkeitsschemas. In: *Arbeiten zur Informationstheorie I*. Berlin: Deutscher Verlag der Wissenschaften, pp. 85–90.
- [21] Fishburn, P. C. (1970). *Utility Theory for Decision Making*. New York: Wiley.
- [22] Gleason, A. M. (1957). Measures on the closed subspaces of a Hilbert space. *J. Math. Mech.* 6: 885–893.
- [23] Gneiting, T., Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *J. Amer. Statist. Assoc.* 102(477): 359–378.
- [24] Grünwald, P. D., Dawid, A. P. (2004). Game theory, maximum entropy, minimum discrepancy and robust Bayesian decision theory. *Ann. Statist.* 32(4): 1367–1433.
- [25] Hamel, G. (1905). Eine Basis aller Zahlen und die unstetigen Lösungen der Funktionalgleichung  $f(x + y) = f(x) + f(y)$ . *Math. Ann.* 60: 459–462.
- [26] Herstein, I. N., Milnor, J. (1953). An axiomatic approach to measurable utility. *Econometrica*. 21: 291–297.
- [27] Holevo, A. S. (1982). *Probabilistic and Statistical Aspects of Quantum Theory*. Amsterdam: North-Holland.
- [28] Jensen, J. L. W. V. (1906). Sur les fonctions convexes et les inégalités entre les valeurs moyennes. *Acta Math.* 30: 175–193.
- [29] Kendall, D. G. (1964). Functional equations in information theory. *Z. Wahrsch. Verw. Gebiete*. 2: 225–229.
- [30] Kestelman, H. (1947). On the functional equation  $f(x + y) = f(x) + f(y)$ . *Fund. Math.* 34: 144–147.
- [31] Khinchin, A. I. (1957). *Mathematical Foundations of Information Theory*. New York: Dover.
- [32] Koller, D., Friedman, N. (2009). *Probabilistic Graphical Models: Principles and Techniques*. Cambridge, MA: MIT Press.
- [33] Kormes, M. (1926). On the functional equation  $f(x + y) = f(x) + f(y)$ . *Bull. Amer. Math. Soc.* 32: 689–693.
- [34] Kreps, D. M. (1988). *Notes on the Theory of Choice*. Boulder, CO: Westview Press.
- [35] Kuczma, M. (2009). *An Introduction to the Theory of Functional Equations and Inequalities: Cauchy’s Equation and Jensen’s Inequality*, 2nd ed. (A. Gilányi, ed.). Basel: Birkhäuser.
- [36] Lee, P. M. (1964). On the axioms of information theory. *Ann. Math. Statist.* 35: 415–418.

- [37] Lyapunov, A. A. (1940). Sur les fonctions-vecteurs complètement additives. *Izv. Akad. Nauk SSSR Ser. Mat.* 4: 465–478. (Accessible modern exposition: Halmos, P. R. (1948). The range of a vector measure. *Bull. Amer. Math. Soc.* 54: 416–421.)
- [38] McConway, K. J. (1981). Marginalization and linear opinion pools. *J. Amer. Statist. Assoc.* 76(374): 410–414.
- [39] Murphy, A. H. (1973). A new vector partition of the probability score. *J. Appl. Meteorol.* 12: 595–600.
- [40] Ostrowski, A. M. (1929). Über die Funktionalgleichung der Exponentialfunktion und verwandte Funktionalgleichungen. *Jahresber. Deutsch. Math.-Verein.* 38: 54–62.
- [41] Reem, D. (2017). Remarks on the Cauchy functional equation and variations of it. *Aequationes Math.* 91(2): 237–264. doi.org/10.1007/s00010-016-0463-6
- [42] Reid, M. D., Williamson, R. C. (2011). Information, divergence and risk for binary experiments. *J. Mach. Learn. Res.* 12: 731–817.
- [43] Savage, L. J. (1971). Elicitation of personal probabilities and expectations. *J. Amer. Statist. Assoc.* 66(336): 783–801.
- [44] Schervish, M. J. (1989). A general method for comparing probability assessors. *Ann. Statist.* 17(4): 1856–1879. doi.org/10.1214/aos/1176347398
- [45] Sierpiński, W. (1920). Sur les fonctions convexes mesurables. *Fund. Math.* 1: 125–128.
- [46] Sierpiński, W. (1922). Sur les fonctions d'ensemble additives et continues. *Fund. Math.* 3: 240–246.
- [47] Solovay, R. M. (1970). A model of set-theory in which every set of reals is Lebesgue measurable. *Ann. of Math.* (2). 92(1): 1–56. doi.org/10.2307/1970696
- [48] Steinhaus, H. (1920). Sur les distances des points dans les ensembles de mesure positive. *Fund. Math.* 1: 93–104.
- [49] Steinwart, I. (2007). How to compare different loss functions and their risks. *Constr. Approx.* 26(2): 225–287. doi.org/10.1007/s00365-006-0662-3
- [50] Tewari, A., Bartlett, P. L. (2007). On the consistency of multiclass classification methods. *J. Mach. Learn. Res.* 8: 1007–1025.
- [51] Tverberg, H. (1958). A new derivation of the information function. *Math. Scand.* 6: 297–298.
- [52] von Neumann, J., Morgenstern, O. (1947). *Theory of Games and Economic Behavior*, 2nd ed. Princeton, NJ: Princeton Univ. Press. (First ed. 1944; the expected-utility axiomatization appears in the appendix added to the 2nd edition.)
- [53] Wright, V. J., Weigert, S. (2019). Gleason-type theorems from Cauchy's functional equation. *Found. Phys.* 49(6): 594–606. doi.org/10.1007/s10701-019-00275-x
- [54] Wright, V. J., Weigert, S. (2020). Correction to: Gleason-type theorems from Cauchy's functional equation. *Found. Phys.* 50(5): 511–514. doi.org/10.1007/s10701-020-00340-w